



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No: 03

Team No: 15

Project Title: Linux Inter-Process Communication Using Pipes and FIFOs

Team Members

Roll Number	Student Name	Signature
2520030471	SAI ANANYA ROY	
2520030504	DEVUNI SHREENITHA	

1. Abstract

The **Hospital Emergency Patient Monitoring System** is a Linux-based project that demonstrates **Inter-Process Communication (IPC)**. It simulates patient monitoring, emergency processing, and doctor alerts using multiple processes. The system takes the **Patient ID and heart rate**, sends the data through a pipe, checks the patient's status, and sends the result through a FIFO. The system also displays the **simulated condition, recommended action, and doctor alert**.

2. Problem Statement

To show how different Linux processes can **communicate and work together** to receive patient information, process the heart rate, and generate an emergency alert using pipes and FIFOs.

3. Objectives

- * To simulate a hospital emergency monitoring system.
- * To demonstrate **Inter-Process Communication (IPC)**.

- * To use **pipes and FIFOs** for process communication.
- * To transfer and process **Patient ID and heart rate**.
- * To demonstrate **process creation and synchronization**.

4. Proposed Methodology

* **Patient Monitoring** – Takes the Patient ID and heart rate from the user and sends the data through a pipe.

* **Emergency Processor** – Receives the patient data, checks the heart rate, and gives a status such as **LOW, NORMAL, WARNING, or EMERGENCY**. It also displays the simulated condition and recommended action.

* **Doctor Alert** – Receives the processed information through a **FIFO** and displays the emergency alert.

Flow:

Patient Monitoring → Pipe → Emergency Processor → FIFO → Doctor Alert

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
fork()	Creates processes
pipe()	Communication between related processes
mkfifo()	Creates a named pipe
read()	Receives data
write()	Sends data
open() / close()	Manages communication channels
waitpid()	Process synchronization
IPC	Enables process communication
Process Synchronization	Coordinates processes
File Descriptors	Accesses pipes and FIFOs

execl()	Executes the Doctor Alert program
unlink()	Removes the FIFO after use

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030471	SAI ANANYA ROY	Abstract
2520030504	DEVUNI SHREENITHA	Code Implementation

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Project execution environment
C / C++	Program implementation
GCC	Compilation
POSIX APIs	System and IPC functions
VS Code/	Compilation and testing
Linux Terminal	Code development

8. Expected Outcome

The system will transfer patient information between multiple Linux processes, classify the patient's simulated status, and generate an alert through IPC. The project demonstrates practical use of processes, pipes, FIFOs, synchronization, and Linux system calls.

9. GitHub Repository

GitHub Repository Link: https://github.com/shreenithadevuni/Linux-Inter-Process-Communication-Using-Pipes-and-FIFOs-_OSSP

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____